Cornell University's Information Science (IS) PhD program offers a unique opportunity to combine my background in human-computer interaction (HCI) and social science with the computational knowledge of information systems necessary to design, build, and evaluate AI-mediated communication systems. As social interaction is increasingly mediated by algorithms, I envision a future in which community-driven intelligent systems understand human intent and learn from feedback to personalize interactions that empower human-human communication and collaboration on the internet. To this end, I am interested in (1) investigating informal learning and community development on online platforms and (2) designing and building human-in-the-loop AI systems that support these processes. After graduation, I plan to pursue tenure-track professorships in human-AI interaction, computational social science, and social computing.

Online platforms such as Reddit and YouTube increasingly support distributed collaborative processes in fields such as education, healthcare, and work. Despite the increasing ubiquity of algorithms that augment communication, significant challenges remain combining large-scale approaches common in machine learning (ML) with human-centered perspectives in social computing to support labor intensive and value sensitive social interaction. For example, classification algorithms for hate speech trained without context awareness can exacerbate pre-existing biases that are damaging to smaller communities on online platforms, despite being intended to prevent harm and reduce manual labor [1]. These problems motivate me to conduct research on human-AI interaction theory and sociotechnical systems to understand and shape AI's impact on society.

My experience in social computing, computational social science, and technical HCI has prepared me to tackle this research direction. Under the tutelage of Dr. Geoff Kaufman and Dr. Joseph Seering during a Master of Human-Computer Interaction at Carnegie Mellon University (CMU), I conducted semi-structured interviews and analyzed data using grounded theory to help develop a cross-platform theory on how online community development is impacted by human moderation. We found that moderators play a key role in the growth of online groups by monitoring antisocial behavior and facilitating discussion, but have a complex decision making process for handling violations manually that includes instructing offenders on community norms [2]. These findings elucidate a need to support community growth and values in automated governance. After graduating from CMU, I worked as a User Experience Researcher at Google leveraging methods such as diary studies, usability testing, and large-scale surveys to inform interface design for content on Google Maps. Combined, these experiences have provided me with rigorous training in mixed-methods research.

To gain experience with ML algorithms, I left Google to pursue a MS in CS at the Georgia Institute of Technology. Advised by Dr. Diyi Yang at Stanford University, my recent research investigates how conversations with non-professional volunteer counselors impact mental health support seekers on an online therapy platform. I led a collaboration with two other Master's students to develop a data science pipeline that includes data collection from a SQL database, feature engineering using computational linguistics techniques in Python, and regression modeling in R. We found that a single high-quality counseling conversation shows no significant impact on support seeker mood and may even lead to seekers dropping off the platform. As first author on a paper currently in submission to CHI 2023, I argue that better individual outcomes may be at odds with community growth outcomes based on a triangulation of multiple metrics [3]. Since counselors do not always receive feedback from those they have helped, designing alternate metrics and tools for counseling impact can help platforms retain counselors as vital parts of their community.

To gain experience in the engineering of sociotechnical systems, I have also worked with Dr. Haijun Xia at the University of California San Diego's Design Lab to develop a videoconferencing system that extracts intent from real-time speech, retrieves information relevant to ongoing discussion, and intelligently suggests interface actions [4]. My core contribution is the design of novel language-oriented interactions. I identified implicit and explicit cues that people use in remote meetings to direct the attention of others through a videography study of recorded meetings. To build the prototype, I also created interface mockups in Figma, implemented a semantic search engine in Python using Transformers, and led user testing with 17 participants. While language model capabilities have expanded in recent years, designing real-time interactions remains a challenge for

mixed-initiative language user interfaces. This project has prepared me for research in technical HCI with hands-on experience in systems prototyping.

Cornell provides an unrivaled chance to pursue research integrating social and computational perspectives in AI-mediated communication and its impact on society. I am interested in working together with Dr. Mor Naaman to develop systems that support human values in governance and communication for social and political discourse online. Dr. Qian Yang's research investigating human-AI interaction design theory is an area I wish to gain further expertise in, collaborating together on addressing ways to improve feedback and control of AI tools such as large language models. Studying computational social science methods with Dr. Cristian Danescu-Niculescu-Mizil would allow me to develop further expertise in analyzing conversational behavior for moderation and counseling at scale. Dr. Malte Jung's work with social robots parallels my interests in human-in-the-loop AI, and I look forward to opportunities to learn more about how to use intelligent agents to moderate human-human collaboration in distributed teams.

My non-traditional path to computing has given me an interdisciplinary, human-centered approach to not only scholarship, but also pedagogy. I currently teach a masters-level Introduction to HCI course to students with diverse backgrounds as an Adjunct Assistant Professor of Computer Science at Pace University, where I regularly use user-centered design methods to collect feedback from students and improve the course syllabus. I look forward to training human-centered scientists, designers, and engineers as a full-time professor after a PhD, shaping a generation of individuals addressing an increasingly AI-mediated future.

References

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